

CAT 2025

MBA Fastrack



Lecture - 4

ALGEBRA

Logarithm

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#Q. The number of solutions of the equation

$$4^x - 3(2)^x - 4 = 0$$

are

$$2^x = t$$

A

0

$$t^2 - 3t - 4 = 0$$

$$t^2 = (2^x)^2 = (2^2)^x = 4^x$$

B

1 ✓

$$(t - 4)(t + 1) = 0$$

C

2

$$t = 4 \quad t = -1$$

$$2^x = 4 \quad 2^x = -1^x$$

D

3

$$\boxed{x = 2} \checkmark$$

TOPICS *to be covered*

- 1 Logarithm
- 2 Cubic Equation



Topics : Logarithm

LOGARITHM MEANING



The logarithm of a number to a given base is the exponent to which that base must be raised in order to produce the number.

$$\log_b x = y$$

“ ‘b’ KI POWER KYA KAROGI KI ‘x’ MILEGA ”

$$b^y = x$$

$$\log_2 8 = 3$$

$$\log_3 81 = y$$

“ ‘3’ KI POWER KYA KAROGI KI ‘81’ MILEGA ”

$$3^y = 81$$

$$3^y = 3^4$$

$$y = 4$$

□ Logarithm are nothing but reverse of exponents.

Let's Get Comfortable



$$(i) \log_5 25 = 2$$

$$(iv) \log_5 1 = 0$$

$$\log_N 1 = 0$$

$$(ii) \log_{1000} 10 = x$$

BASE $\rightarrow$ POWER

$$1000^x = 10$$

$$10^{3x} = 10^1$$
$$3x = 1$$
$$x = \frac{1}{3}$$

$$(v) \log_5 1/5 = -1$$

$$\log_N \frac{1}{N} = -1$$

COMMON LOG $\rightarrow$ BASE = 10 ✓

$$(iii) \log 100$$

$$\log_{10} 100 = 2$$

$$(vi) \log_5 5 = 1$$

$$\log_N N = 1$$



Logarithm – Properties + Questions

QUESTION-1



#Q. Solve for x

$$\log_3(x+1) = 2$$

Handwritten annotations: A bracket above the argument $(x+1)$ and a curved arrow pointing from the base 3 to the argument, with the text "Ki power" written below the arrow.

"Single log.
Convert into
exponents"

$$3^2 = x+1$$

$$9 = x+1$$

$$x = 8$$

A 7

B 6

C 5

D 8 ✓✓

QUESTION-2



#Q. Find the sum of all the possible values of x which satisfy the following equation.

$$\log_x(5x + 14) = 2$$

A

7 ✓✓

B

6

C

5 ✗

D

8

$$x^2 = 5x + 14$$

$$x^2 - 5x - 14 = 0$$

$$(x - 7)(x + 2) = 0$$

$$\checkmark \boxed{x = 7}$$

$$\boxed{x = -2}$$

$$\boxed{\text{base} < 0}$$

✗
∴

$$\log_b x$$

$$\boxed{N > 0} \checkmark$$

$$\boxed{b > 0}$$

$$\boxed{b \neq 1}$$



QUESTION-3



#Q. Find x

$$\log_2(\log_2(x-1)) = 2$$

A

17 ✓

B

13

C

65

D

19

$$\log_2(x-1) = 4$$

$$x-1 = 16$$

$$x = 17$$

~~$$\log_2(\log_2(x-1)) = 2$$~~

16

$$x-1 = 16$$

$$x = 17$$

QUESTION-4



#Q. Find x

$$\log_2[5 - \log_2(x - 1)] - 2 = 0$$

A

7

B

3 ✓

C

5

D

9

~~$\log_2[5 - \log_2(x - 1)] = 2$~~

Handwritten work showing the solution process:

$\log_2[5 - \log_2(x - 1)] = 2$

$5 - \log_2(x - 1) = 4$

$-\log_2(x - 1) = 4 - 5$

$-\log_2(x - 1) = -1$

$\log_2(x - 1) = 1$

$x - 1 = 2$

$x = 3$

QUESTION-5 (CAT 2021 Slot 2)



#Q. If $\log_2[3 + \log_3\{4 + \log_4(x - 1)\}] - 2 = 0$ then 4x equals

$$x - 1 = \frac{1}{4}$$

$$x = 1 + \frac{1}{4}$$

$$x = \frac{5}{4}$$

$$\boxed{4x = 5} \checkmark$$

QUESTION-6 (CAT 2023 Slot 1)



#Q. If x and y are positive real numbers such that $\log_x(x^2 + 12) = 4$ and $3\log_y x = 1$, then $x + y$ equals

A

20

B

11

C

68

D

10 ✓

20

$x^2 = t$
 $x^4 = t^2$ → positive

$$x^4 = x^2 + 12$$

$$t^2 = t + 12$$

$$t^2 - t - 12 = 0$$

$$(t-4)(t+3) = 0$$

$$t = 4 \quad t = -3$$

$$x^2 = 4$$

$$x = +2 \text{ or } x = -2$$

$$x = 2$$

$$y = 8$$

$$\log_y 2^3$$

$$3\log_y 2 = 1$$

$$\log_y 2 = \frac{1}{3}$$

$$y^{1/3} = 2$$

$$y = 2^3 = 8$$

$$\log_y 2^3 = 1$$

$$8 = y$$



logarithmic Properties ||

- $\log_{\underline{N}} a + \log_{\underline{N}} b = \log_N ab$

$$\log_4 2 + \log_4 8 = \log_4 16 = 2$$

- $\log_N a - \log_N b = \log_N \frac{a}{b}$

$$\log_3 12 - \log_3 4$$

$$\log_3 3 = 1$$

$$a^x \cdot a^y = a^{x+y}$$

$$\frac{a^x}{a^y} = a^{x-y}$$

QUESTION-7



#Q. Solve

$$\log_a ab^2 + \log_a c - \log_a b^2c =$$

A 1 ✓

B 2

C 3

D 4

$$\log_a \left[\frac{ab^2c}{b^2c} \right]$$

$$\log_a a = 1$$

$$\log_7 63 \longrightarrow \log_7 (7 \times 9)$$

$$\hookrightarrow \log_7 7 + \log_7 9$$

$$1 + \log_7 9$$

$$(A) 1 + \log_7 3$$

$$\checkmark (B) 1 + \log_7 9$$

$$(C) 1 - \log_7 3$$

$$(D) 1 - \log_7 9$$

Logarithmic Properties |||

24

$$\log_5 120$$

$$\log_5 (5 \times 4 \times 6)$$

$$\log_5 5 + \log_5 4 + \log_5 6$$

$$1 + \log_5 4 + \log_5 6$$

TAGDI PROPERTY - I



- $\log_N a^k = k \log_N a$

$$\log_7 3^4 = 4 \log_7 3$$

$$\log_3 81 = 4$$

$$\log_3 3^4$$

$$4 \log_3 3 = 4(1) = 4$$

- $\log_{N^k} a = \frac{1}{k} \log_N a$

$$\log_{3^4} 3 = \frac{1}{4} \log_3 3$$

Reciprocal

$$= \frac{1}{4} (1) = \frac{1}{4}$$

As it is

$$\log_{3^{\frac{7}{2}}} 3^{\frac{7}{2}}$$

Reciprocal

$$= \frac{7}{2} (\cancel{2}) \log_3 3$$

$$= 7(1)$$

$$= 7$$

QUESTION - 8



#Q. Find $N + M$

$$N = \log_7 \sqrt[5]{a^3} - \log_{a^{\frac{1}{6}}} a^{\frac{1}{5}}$$

$$M = \log_{0.008} 125$$

A 1

B 2 ✓✓

C 3

D 4

$$\log a^{\frac{3}{5}} - \left(\frac{1}{5}\right)(6)$$

$$3\left(\frac{7}{5}\right) - \frac{6}{5}$$

$$\frac{21}{5} - \frac{6}{5}$$

$$\frac{15}{5}$$

$$3$$

$$\log_{\frac{1}{125}} 125 = -1$$

$$0.008$$

$$\frac{8}{1000}$$

$$\frac{1}{125}$$

QUESTION-9



#Q. Solve

Given $\log_5(2x^3 - 4x^2) = 3 \log_5 x$, find the product of all the possible values of x .

A 0

B 2

C 1

D 4 ✓

$$\log_5(2x^3 - 4x^2) = 3 \log_5 x$$

$$\log_5(2x^3 - 4x^2) = \log_5 x^3$$

$$2x^3 - 4x^2 = x^3$$

$$x^3 = 4x^2$$

$$\boxed{x = 4}$$

TAGDI PROPERTY - 2



Base Change theorem

$$\log_{27} 81$$
$$\frac{\log_3 81}{\log_3 27} = \frac{4}{3}$$
$$\log_{3^3} 3^4$$
$$(4)\left(\frac{1}{3}\right)$$
$$\frac{4}{3}$$

$$\frac{\log_7 (2^x + 3)}{\log_{7^2} 2^x}$$
$$\frac{\log_7 (2^x + 3)}{\frac{1}{2} \log_7 2^x}$$
$$2 \left[\frac{\log_7 (2^x + 3)}{\log_7 2^x} \right] = 2 \log_{2^x} (2^x + 3)$$

QUESTION-10



#Q. **Solve**

$$\log_a b \cdot \log_b c \cdot \log_c a =$$

- A** 0
- B** 1 ✓✓
- C** 2
- D** abc

$$\left(\frac{\log b}{\log a} \right) \left(\frac{\log c}{\log b} \right) \left(\frac{\log a}{\log c} \right)$$



logarithmic Property V

TAGDI PROPERTY - 2



- $a^{\log_a b} = b$

$$\frac{(5 \times 2)}{2}$$

$$5^{\log_5 x} = x$$

$$\left\{ \begin{array}{l} \underline{3}^{\log_3 9} \\ 3^2 \\ 9 \end{array} \right\} \longrightarrow 9$$

QUESTION-11



#Q. **Solve**

$$\frac{5^{\log_5 x^3} - 1}{3^{\log_3 x} - 2 \log_a \sqrt{a}}$$

- A** $x + 1$
- B** $x^2 + x + 1$ ✓✓
- C** $x^2 - x + 1$
- D** $x - 1$

$$\frac{x^3 - 1}{x - 2 \log_a a^{1/2}}$$

$$\frac{(x-1)(x^2+x+1)}{x - 2\left(\frac{1}{2}\right)[\log_a a]} = \frac{(x-1)(x^2+x+1)}{x-1}$$

QUESTION-12 (CAT 2024 – SLOT 2)



#Q. If a, b and c are positive real number such that $a > 10 \geq b \geq c$ and

$$\frac{\log_8(a+b)}{\log_2 c} + \frac{\log_{27}(a-b)}{\log_3 c} = \frac{2}{3}$$

then the greatest possible integer value of a is

$$\frac{\frac{1}{3} \log_2(a+b)}{\log_2 c} + \frac{\frac{1}{3} \log_3(a-b)}{\log_3 c}$$

$$\frac{1}{3} \left[\log_c(a+b) \right] + \frac{1}{3} \left[\log_c(a-b) \right] = \frac{2}{3}$$

$$\log_c(a+b)(a-b) = 2$$

$$\log_c(a^2 - b^2) = 2$$

$$a^2 - b^2 = c^2$$

$$a^2 = b^2 + c^2$$

$$a^2 = 10^2 + 10^2$$

(MAX)

$$a^2 = 200$$

(MAX)

$$a = 10\sqrt{2}$$

(MAX)

$$a = 10(1.414)$$

(MAX)

$$= 14.14$$

Int $\rightarrow 14$ ✓

QUESTION-13



#Q. Solve

$$(3^2)^3 = 3^6$$

Let a , b , and c be three positive real numbers such that $\log_5 a^2 + \log_{25} b^{3/2} + \log_{125} c^{4/3} = 12$. Find the value of $a^{12} b^9 c^8$?

A 1

B 25

C 5^{72} ✓

D 72

$$\log_5 a^2 + \frac{1}{2} \log_5 b^3 + \frac{1}{3} \log_5 c^4 = 12$$

$$\log_5 a^2 + \log_5 (b^3)^{1/2} + \log_5 (c^4)^{1/3} = 12$$

$$\log_5 a^2 b^{3/2} c^{4/3} = 12$$

$$(5^{12})^6 = (a^2 b^{3/2} c^{4/3})^6$$

QUESTION-14 (CAT 2024 – SLOT 1)



#Q. If x is a positive real number such that

$$4 \log_{10} x + \frac{4}{2} \log_{10^2} x + \frac{8}{3} \log_{10^3} x = 13$$

then the greatest integer not exceeding x is

$$4 \log_{10} x + 2 \log_{10} x + \frac{8}{3} \log_{10} x = 13$$

$$\left[4 + 2 + \frac{8}{3}\right] \log_{10} x = 13$$

$$\frac{26}{3} \log_{10} x = 13$$

$$\log_{10} x = \frac{3}{2}$$

$$x = 10^{3/2}$$

$$x = \sqrt{10^3}$$

$$x = \sqrt{1000}$$

$$x = 31.6$$

31 ✓

32

SQ. 1 to 100

$$\begin{cases} 30^2 = 900 \\ 31^2 = 961 \\ 32^2 = 1024 \end{cases}$$



Topics : Cubic Equation

$$\blacksquare \quad x^3 - 6x^2 + 11x - 6 = 0 \quad \begin{matrix} \nearrow 1 \\ \rightarrow \alpha \\ \searrow \beta \end{matrix}$$

$H \notin T$	$1^3 - 6(1)^2 + 11(1) - 6$
$x = 1, 2, 3$	$1 - 6 + 11 - 6$
$-1, -2, -3$	$12 - 12 = 0$

$$\text{Sum} = -\frac{b}{a}$$

$$\text{Prod} = -\frac{d}{a}$$

$$1 + \alpha + \beta = -\frac{(-6)}{1}$$

$$(1)(\alpha)(\beta) = -\frac{(-6)}{1}$$

$$\alpha + \beta = 6 - 1$$

$$\boxed{\alpha\beta = 6}$$

$$\boxed{\alpha + \beta = 5}$$

$$\alpha = 2 \quad \beta = 3$$

$$\blacksquare \quad x^3 - 12x^2 + 39x - 28 = 0$$

$$(1, \alpha, \beta)$$

$$1 + \alpha + \beta = 12$$

$$(1)(\alpha)(\beta) = 28$$

$$\alpha + \beta = 11$$

$$\alpha\beta = 28$$

$$\alpha = 4$$

$$\beta = 7$$



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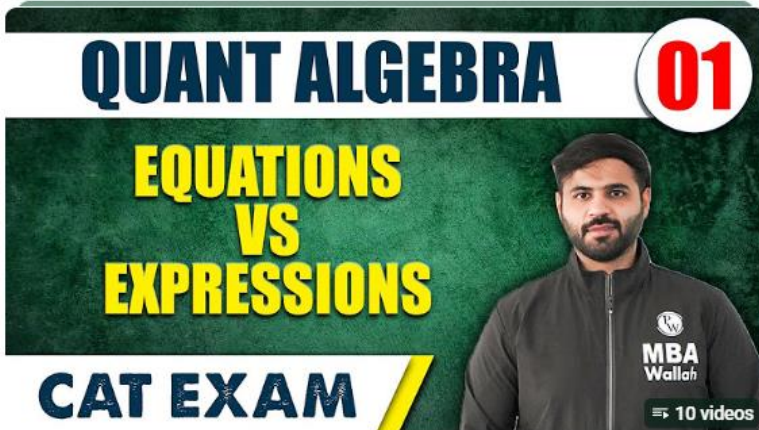
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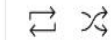


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